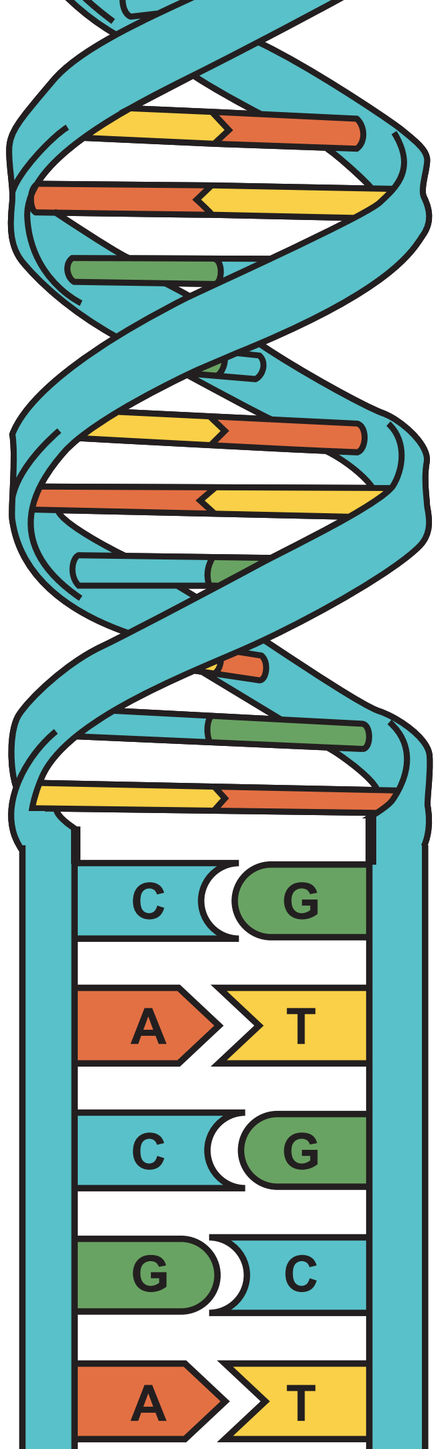


STRUCTURE AND FUNCTION OF D-DEOXYRIBO N-NUCLEIC A-ACID

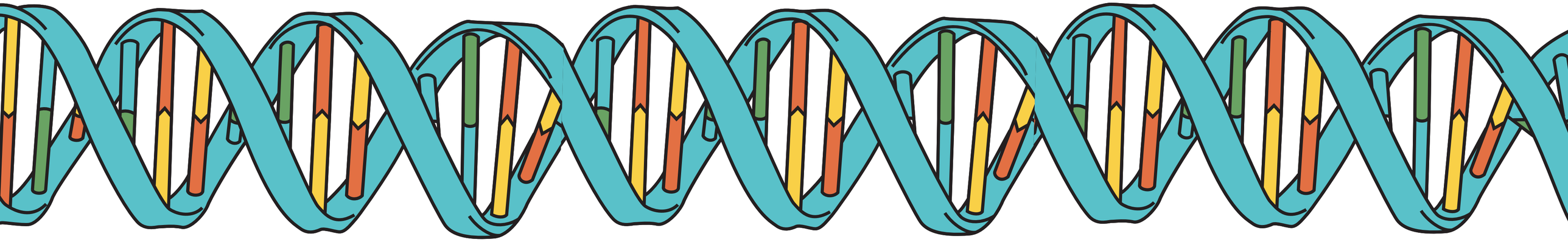
BY: SELLAR, APRIL R.

WHAT IS DNA?

DNA, or deoxyribonucleic acid, is the molecule responsible for storing and transmitting genetic information in all living organisms. Found in the nucleus of cells, DNA provides the instructions that guide the development, functioning, and reproduction of all life. This paper explores the structure of DNA, the base-pairing rules, and its vital role in passing traits from one generation to the next.

THE DOUBLE HELIX MODEL

The structure of DNA was first described by James Watson and Francis Crick in 1953. They discovered that DNA has a double helix shape, which looks like a twisted ladder.



Backbone:

The sides of the ladder are made of alternating sugar (deoxyribose) and phosphate groups.

Rungs:

The steps of the ladder are pairs of nitrogenous bases connected in the middle. The two strands run in opposite directions (antiparallel) and twist around each other, forming the stable helical shape.

DNA IS MADE OF FOUR NITROGENOUS BASES:

* ADENINE (A)
* THYMINE (T)
* CYTOSINE (C)
* GUANINE (G)

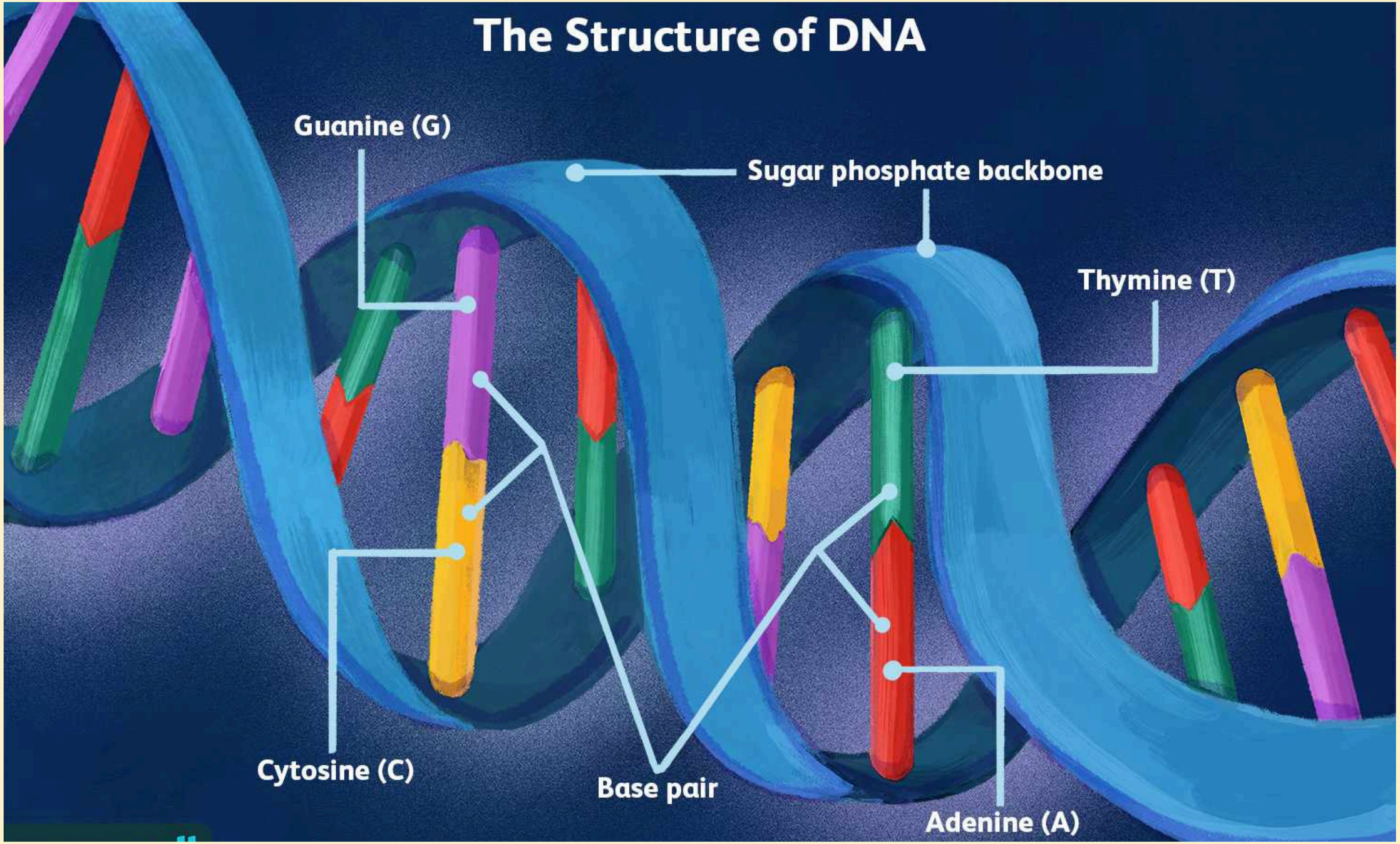
THE BASE-PAIRING RULES ARE:

* **ADENINE (A)** ALWAYS PAIRS WITH **THYMINE (T)**
* **CYTOSINE (C)** ALWAYS PAIRS WITH **GUANINE (G)**

THESE SPECIFIC PAIRINGS ARE HELD TOGETHER BY HYDROGEN BONDS:

* A-T PAIRS HAVE 2 HYDROGEN BONDS
* C-G PAIRS HAVE 3 HYDROGEN BONDS

THIS BASE-PAIRING ENSURES ACCURACY DURING DNA REPLICATION AND HELPS PRESERVE GENETIC INFORMATION.


The Structure of DNA
Guanine (G)
Sugar phosphate backbone
Thymine (T)
Cytosine (C)
Base pair
Adenine (A)

UNDERSTANDING DNA IS IMPORTANT IN MANY FIELDS:

Real-life Application

FORENSICS: DNA FINGERPRINTING HELPS SOLVE CRIMES.

ANCESTRY TESTING: DNA REVEALS YOUR ETHNIC ROOTS AND FAMILY HISTORY.

MEDICINE: DOCTORS USE DNA TESTS TO DETECT GENETIC DISORDERS.

BIOTECHNOLOGY: SCIENTISTS USE DNA KNOWLEDGE IN GENE THERAPY AND VACCINE DEVELOPMENT.

CONCLUSION

DNA IS THE BLUEPRINT OF LIFE. ITS DOUBLE-HELIX STRUCTURE AND BASE-PAIRING RULES ALLOW IT TO CARRY AND PROTECT GENETIC INFORMATION. THROUGH INHERITANCE, DNA CONNECTS GENERATIONS AND DETERMINES TRAITS. UNDERSTANDING DNA HAS LED TO MAJOR BREAKTHROUGHS IN SCIENCE, MEDICINE, AND TECHNOLOGY.

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THANK YOU!!!